

Gear Wall Element B

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Attempt	Prior Attempts	Successful/Unsuccessful	Strengths	Weaknesses
Bubble Tube	During a multi sensory environment study involving 41 autistic children were put into a sensory room with the following implements: Bubble tube, Touch sound and Light Board, Tactile Board and Fiber Optic Stimulation Display. In seeking to find which stimulation device is most effective this study compared how the students interacted with the devices including information on how often the devices were picked and how long they kept the students occupied.	Was shown to be highly successful during the study as it received the second highest mean visits falling just below the touch light and sound board while having the longest average engagement time.	Provides the students with both auditory and visual stimulation	Does not provide tactile stimulation. This may make it more difficult to use it for cause and effect learning and motor skill training.
Touch sound and Light Board		The touch sound and Light Board was effective in engaging children. Also in the study, the touch sound and Light Board Were the highest in the frequencies of visits and second in the duration of the visit, but the success of the device did depend on the individual's cognitive abilities and intellectuality.	Catering to more diverse sensory needs with it combining tactile, auditory, and visual elements. Also it worked well in capturing the child's attention while also being flexible to the child's sensory profile and intellectual abilities.	Children with specific sensory sensitivities to lights or noises, may be overwhelmed or harmed with all the different types of sensory capabilities of the device
Tactile Board		This device was not successful compared to the other devices because it had the least amount of visits and the duration of the visit to the device was the lowest at 20-30 seconds, while the other device lasted about a minute or more.	Offers various textures to simulate sensory exploration, also it promotes active engagement to the children.	Less effective for children without strong tactile preferences and provides tactile input only, lacking the multisensory integration of other devices.
Fiber Optic Stimulation Display		While this device proved to be less entertaining to the children during the study it still captivated them more than the tactile board with both a higher mean visit frequency and an average visit time of over 40 seconds.	Strengths: Visually captivating: Provides dynamic light displays, effectively engaging children with visual sensory preferences. Calming effect: Can help reduce anxiety and promote relaxation in sensory-sensitive individuals.	Weaknesses: Limited tactile interaction: Primarily focused on visual stimuli, offering little for those who benefit more from hands-on engagement. Potential overstimulation: Bright lights and colors may overwhelm some users, particularly those sensitive to intense visual input.

Light Wall	https://www.youtube.com/watch?v=T9j6rQ4rtQY In Connecticut, Hanover Elementary School installed a sensory room to provide their students with special needs with personalized sensory input in order to help them to be better prepared for learning and interacting with others.	The room has had a positive impact on students because they are better able to follow directions and spend more time on tasks.	The wall offers visual stimulation, development of motor skills, and individualized use.	The light wall might not fully meet the needs of students who require other types of sensory input and may cause overstimulation due to the abundance of lights.
Digital Touch Board	https://www.youtube.com/watch?v=PxrpaJROER8&t=18s https://www.youtube.com/watch?v=JPoybQOL8Us&t=6s In Delaware, New Castle County School installed a sensory room which involved various stimulation devices such as a light up board that responds to touch.(2022)	The touch board was successful in that it allowed the students to interact through touch and sound which fostered a cause and effect learning for these students.	Gives visual stimulation and cause and effect learning which allows for understanding of their action.	The board requires many sensors which could break and would require the board to be replaced.

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